

WearUS: A Unified Framework for Wearable A-Mode Ultrasound Pretraining and Downstream Transfer

Eleventh Weekly Report

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May 27, 2026

Quantitative Results of Pretraining per Dataset

Dataset	RMSE	R	R ²	Pretrained	Channels
braush_contraction	0.009027	0.986435	0.973019	Yes	1
braush_fatigue	0.021547	0.971385	0.943547	Yes	1
giordano_heartrate	0.010344	0.987507	0.974964	Yes	1
grawus	0.022009	0.992160	0.984380	Yes	6
hwc	0.018396	0.996099	0.992212	Yes	4
lat_gastrocnemius_veras.	0.031491	0.896125	0.802972	Yes	196
picmus_carotid_cross	0.043527	0.691357	0.477760	Yes	64
picmus_carotid_long	0.042545	0.695391	0.483426	Yes	64
picmus_in_vivo_heart	0.039231	0.718081	0.514324	Yes	128
SpaconeBicepsTricipes	0.100359	0.710844	0.496401	No	6

Experimental Setup (Single Patient Focus):

- **Dataset:** Not balanced, focused on a **single patient**, including all available sessions.
- **Data Split Strategy:**
 - **Train/Val:** 5 sessions (70% / 30% random split).
 - **Test:** 6th independent remaining session.

Performance Comparison:

Model	Train Acc	Acc (10Ep)	Loss (10Ep)	Acc (1Ep)	Loss (1Ep)
pretrained_FixedS	99%	61%	2.65	61%	1.27
pretrained_interp	99%	79%	1.39	75%	0.75
from scratch	99%	17%	7.26	-	-

Epochs	Train Accuracy	Test Accuracy	Test Loss
1	90%	60%	1.98
10	99%	65%	3.23
20	99%	71%	2.25
50	99%	59%	3.76

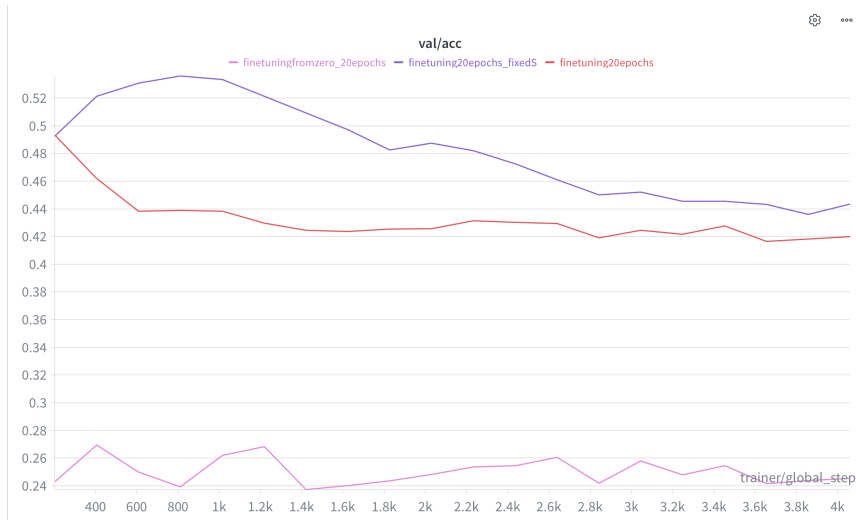
Experimental Setup:

- **Dataset:** Balanced, all patients and all data collection sessions.
- **Data Split Strategy:**
 - **Training:** 4 sessions.
 - **Validation:** 1 separate session.
 - **Testing:** 1 additional independent session.

Performance Comparison:

Model	Train Accuracy	Test Accuracy	Test Loss
pretrained_interp	99%	66%	0.66
from scratch	98%	20%	3.94
CNNBaseline	93%	48%	2.10

Validation trend



Validation accuracy over training epochs